

Ziyi Huang

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Research Interest

Execution functions, Precision neuromodulation, Brain dynamics, Developmental cognitive neuroscience

Education

East China Normal University, Cognitive Neuroscience, Ph.D. candidate (expected June 2026)	2021 -
South China Normal University, Developmental and educational psychology, master	2018 - 2021
Shantou University, Finance, Bachelor	2014 - 2018

Publications

1. **Huang, Z.**, & Yin, D. (2025). Common and unique network basis for externally and internally driven flexibility in cognition: from a developmental perspective. *Developmental Cognitive Neuroscience*, 101528.
2. **Huang, Z.**, Gao, W., Wu, Z., Li, G., & Nie, J. (2023). Functional brain activity is highly associated with cortical myelination in neonates. *Cerebral Cortex*, 33(7), 3985-3995.
3. **Huang, Z.**, Wang, Q., Zhou, S., Tang, C., Yi, F., & Nie, J. (2020). Exploring Functional Brain Activity in Neonates: A Resting-state fMRI Study. *Developmental Cognitive Neuroscience*, 45, 100850.
4. Yang, Y., **Huang, Z.**, Yang Y., & Yin, D. (2025). Time-dependent consolidation mechanisms of durable memory in spaced learning. *Communications Biology*, 8(1), 535.
5. Su, K., **Huang, Z.**, Li, Q., Fan, M., Li, T., & Yin, D. (2024). Dissociable functional responses along the posterior-anterior gradient of the frontal and parietal cortices revealed by parametric working memory and training. *Brain Structure and Function*, 229(7), 1681-1696.
6. Zhu, H., **Huang, Z.**, Yang, Y., Su, K., Fan, M., Zou, Y., ... & Yin, D. (2023). Activity flow mapping over probabilistic functional connectivity. *Human Brain Mapping*, 44(2), 341-361.
7. Gao, W., **Huang, Z.**, Ou, W., Tang, X., Lv, W., & Nie, J. (2022). Functional individual variability development of the neonatal brain. *Brain Structure and Function*, 227(6), 2181-2190.
8. Tang, C., **Huang, Z.**, Zhou, S., Wang, Q., Yi, F., & Nie, J. (2019, October). Movie-Watching fMRI Reveals Inter-subject Synchrony Alteration in Functional Brain Activity in ADHD. In *International Workshop on Graph Learning in Medical Imaging* (pp. 104-111). Springer, Cham.
9. Wang, Z., Yang, Y., **Huang, Z.**, Zhao, W., Su, K., Zhu, H., & Yin, D. (2025). Exploring the transmission

of cognitive task information through optimal brain pathways. *PLOS Computational Biology*, 21(3), e1012870.

10. Li, D., Liu, C., **Huang, Z.**, Li, H., Xu, Q., Zhou, B., ... & Xu, X. (2021). Common and Distinct Disruptions of Cortical Surface Morphology Between Autism Spectrum Disorder Children With and Without SHANK3 Deficiency. *Frontiers in Neuroscience*, 15, 751364.

Presentations & Conference talks

1. **Huang, Z.**, & Yin, D. (2025, June). Neurocomputational mechanisms underlying perceptual and response switching. In *Organization for Human Brain Mapping (OHBM) Annual Meeting of 2025*, Brisbane, Australia, Poster.
2. **Huang, Z.**, & Yin, D. (2025, June). Common and unique network basis for externally and internally driven flexibility in cognition. In *Organization for Human Brain Mapping (OHBM) Annual Meeting of 2025*, Brisbane, Australia, Poster.
3. **Huang, Z.**, & Yin, D. (2023, Oct). Functional differentiation of neural flexibility from childhood to adulthood. In *Graduate Forum of 25th National Academic Conference of Psychology*, Chengdu, China, Outstanding Oral Presentation Award.
4. Tang, C., **Huang, Z.**, Zhou, S., Wang, Q., Yi, F., & Nie, J. (2019, October). Movie-Watching fMRI Reveals Inter-subject Synchrony Alteration in Functional Brain Activity in ADHD. In *International Workshop on Graph Learning in Medical Imaging*, Shenzhen, Poster.

Funded Projects

1. *A study on brain dynamics representation of cognitive flexibility*, the National Natural Science Foundation of China (32271096), 2023-2026—Major Contributor
2. *Subproject 34: Developmental trajectories of brain structures and functional networks underlying executive function*, Major Project of “Brain Science and Brain-Inspired Intelligence”, Science and Technology Innovation 2030 Program (2021ZD0200500), 2021–Present —Core member

Research Experience

1. **Cognitive Neuroscience**
 - a) Brain dynamic representation of cognitive flexibility
 - b) Network basis underlying different types of cognitive flexibility
 - c) Cognitive and neural modulation of cognitive flexibility
2. **Developmental Neuroscience**

- a) Development of cognitive flexibility and its functional network basis
- b) Development and functional differentiation of neural flexibility
- c) Functional brain development of neonates
- d) Relationships between brain structure and function in the neonatal brain

3. Neurodevelopmental Disorders

- a) Morphological analysis of the cerebral cortex in children with autism spectrum disorder
- b) Relationships between cortical gray–white matter and brain function in children with Rett syndrome
- c) Machine learning analysis of brain functional abnormalities in children with ADHD

Technical Skills

4. Neuroimaging Data Analysis

- a) Preprocessing and volumetric/morphological analysis of structural MRI data
- b) Preprocessing and analysis of resting-state fMRI, including static and dynamic brain activity and functional connectivity
- c) Preprocessing and analysis of task-based fMRI, including brain activation and task-based functional connectivity
- d) Integrated analysis of resting-state and task-based brain functions
- e) Multimodal integration of brain structure and function

5. Experimental Techniques

- a) Behavioral experiment design and implementation
- b) Cognitive training interventions
- c) Non-invasive brain stimulation techniques, tES or TMS (planned)

6. Software Proficiency

MATLAB, SPSS, SPM, DPABI, Workbench, FreeSurfer, FSL, ANTs

Awards & Honors

1. Outstanding Oral Presentation Award, Chinese Psychological Society, 2023
2. Outstanding Graduate Award, South China Normal University, 2021
3. National Scholarship for Graduate Students, China Ministry of Education, 2020

Research Strengths

Multimodal MRI data analysis, Dynamic brain network analysis, Preprocessing of MRI data in young children